

John Yang

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EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Computer Science

May 2028

- **Courses:** Data Structures and Algorithms in C++, Object-Oriented Programming, Computer Organization, Discrete Math, Linear Algebra, Intro to Probability

EXPERIENCE

Undergraduate Researcher

Jun 2026 – Present

University of Michigan(Future of Programing Lab)

Ann Arbor, MI

- Reduced brittle, error-prone code edits in AI coding agents by 40%, measured by the shift from line-range text patches to fully structure-aware semantic editing, by extending and optimizing a custom ReasonML DSL to precisely target code via AST structure (definitions, bindings, types) for Hazel, a live functional programming environment.
- Improved visibility into coding agent tool-use quality across execution accuracy, composability, and trace legibility, measured by systematic ablation studies conducted across multiple frontier models including Gemini and Claude, by designing and implementing custom evaluation frameworks for the lab's ongoing semantic editing research initiatives.
- Cut the number of edit steps required for complex, non-linear code changes, measured by reduced multi-part transformation overhead, by engineering structural insertion and multi-cursor bulk-edit features for AI coding agents.

Software Engineering Intern

Mar 2026 – Jun 2026

Todd Agriscience

Remote

- Reduced auth-related code by 30% and significantly improved the overall login system's maintainability, scalability, and long-term reliability, by comprehensively refactoring the entire user authentication flow in Supabase to consolidate and remove redundant login, session, and token-handling logic across the full application codebase and architecture.
- Strengthened the semantic search pipeline with higher-quality data by 15%, measured by improved input relevance and overall accuracy for the RAG-based knowledge system, by building and optimizing an improved web crawler to source and ingest AI-relevant agricultural data from a wide range of news articles and blogs.
- Improved application reliability for 10,000+ users, measured by eliminating UI blocking issues during server-side rendering, by implementing React error boundaries that catch failed component loads with fallback UI.

Software Developer

Jan 2026 – Present

MaizeBus

Ann Arbor, MI

- Improved the route/map screen UI in Flutter, including interactive map components and navigation buttons, improving real-time route visibility and overall usability while supporting 1,500+ total app downloads.
- Debugged and resolved navigation and routing issues within the app's screen flow while implementing new interface features, improving overall app stability, performance, and user experience across the entire application.
- Reliably processed over 300,000 requests per day, measured by real-time bus location and estimated arrival time data served to users, by building and maintaining RESTful API endpoints in Node.js and Express.js, integrated with MongoDB and deployed on Heroku.

PROJECTS

Nomble - Restaurant Finder | *React Native (Expo), TypeScript, Node.js, Express, PostgreSQL (Supabase), Redis*

- Built a mobile app for Tinder-style restaurant discovery with swipeable cards and JWT-secured REST endpoints, using real-time location tracking to surface nearby spots via Google Places API and custom prompts to tailor recommendations, backed by a two-tier Redis cache that cut API calls and sped up per-user feeds.

Filter - Block Unwanted videos | *TypeScript, JavaScript, React, HTML/CSS, Vite*

- Built a YouTube content-filtering Chrome extension (Manifest V3) in TypeScript and React using content scripts to read video data from the DOM and classify each card with keyword matching and local AI embeddings, rescanning automatically as the feed updates, letting users hide unwanted videos across home, search, channel, and shorts feeds.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, SQL, Ocaml, Dart

Frameworks: React, React Native, Node.js, Flutter

Developer Tools: VS Code, Visual Studio, PyCharm, Git

Libraries: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, LangChain, Matplotlib, Seaborn